

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT:PERFORM PROGRAMMING PRINCIPLES
UNIT CODE: 0613 451 05A
CAT 1
MODULE 4,TERM 4
TOTAL:50 MARKS
TIME 1 HOURS 30 MIN

NAME.....CDACCREG.....

DATE.....SIGN.....

Instructions

- ✓ *This paper consists of **TEN (10) questions** and carries a total of **50 marks**.*
- ✓ *Answer **ALL** questions in the spaces provided.*
- ✓ *The time allowed for this paper is **1 hour**.*
- ✓ *Write your **Name, Admission/Registration Number** clearly on this sheet before you begin.*
- ✓ *Marks allocated to each question are shown in brackets at the end of the question or sub-question.*
- ✓ *Read each question carefully before answering. Where an example is required, ensure it is relevant and correctly stated.*

QUESTIONS

1. a) Define the term "programming language." (2 marks)

b) Programming languages can be grouped into different categories. Briefly describe **three** of the following categories, giving one example language for each: (5 marks)

- Procedural languages
- Object-oriented languages
- Functional languages

2. Distinguish between **low-level** and **high-level** programming languages, giving one example of each. (4 marks)

3. State **four** criteria that should be considered when selecting a programming language to meet a given user's requirements (4 marks)

4. A hospital wants to develop a system for managing patient records, appointments, and billing. As the systems developer, identify and explain **five** criteria you would use to select an appropriate programming language for this system. (10 marks)

(Relate each criterion directly to the scenario.)

5. a) Define the term "programming paradigm." (2 marks)

6. Explain **three** common programming paradigms, in each case briefly stating how they approach solving a problem. (6 marks)

7. a) Define functional programming. (2 marks)

b) State **two** key characteristics of functional programming (2 marks)

8. a) Define procedural programming. (3 marks)

b) Give **two** advantages of using the procedural paradigm. (2 marks)

9. a) Define object-oriented programming. (2 marks)

b) State **two** core principles of object-oriented programming (2 marks)

10. Distinguish between imperative and declarative programming paradigms, giving **one** example language for each. (4 marks)